

RelSearch version 0.22.1

User Manual

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Chapter 1

Initial setup

1. Ensure that R ($\geq 4.4.0$) is installed. It is available from the [R Development Core Team website](#).
2. Begin an R session.
3. Execute the following command in R to install required packages.

```
install.packages('https://github.com/manabe0322/RelSearch/releases/  
  ↪ download/v0.22.1/RelSearch_0.22.1.zip', repos = NULL, type = 'win.  
  ↪ binary')  
install.packages(c("shiny", "data.table", "dplyr", "DT", "magrittr", "  
  ↪ pedtools", "Rcpp", "ribd", "shinyFeedback", "shinyjs", "shinythemes",  
  ↪ "waiter"))
```

Note:

For readability, the code shown above may be visually wrapped in this manual. When copying the code, please ensure that wrapped lines are combined into a single line before execution.

Chapter 2

Quick guide

1. Execute the following commands in R to launch the software interface.

```
library(RelSearch)
RelSearch()
```

Then, the software interface will open automatically.

The screenshot displays the RelSearch software interface, version 0.21.0. The interface has a blue header bar with the version number and a navigation menu containing 'Load', 'Result', 'Project', 'Database', 'Settings', 'Example files', and 'Manual'. Below the header, there are three main panels for STR, Y-STR, and mtDNA analysis. Each panel has a 'Victim database' section with a 'Load' button and a 'No file selected' status, a 'Reference database' section with a 'Load' button and a 'No file selected' status, and an 'Allele frequencies' section with a 'Load' button and a 'No file selected' status. At the bottom left, there is a large blue 'Analyze' button.

2. Load files from each **"Load"** button.

The screenshot shows the RelSearch web interface with a blue header bar containing the version 'RelSearch ver. 0.21.0' and navigation links: 'Load', 'Result', 'Project', 'Database', 'Settings', 'Example files', and 'Manual'. Below the header, there are three main sections for different marker types: 'STR', 'Y-STR', and 'mtDNA'. Each section contains a 'Victim database' and a 'Reference database' (except for 'STR' which also has an 'Allele frequencies' section). Each database section has a 'Load' button and a text input field with the placeholder 'no file selected'. Orange arrows point to each of these 'Load' buttons. At the bottom of the 'STR' section, there is a blue 'Analyze' button.

Note:

- RelSearch can analyze data for each marker type (i.e., autosomal STR, Y-STR, and mtDNA) separately as well as for all marker types together.
- For autosomal STR analysis, victim, reference, and allele frequency databases are required.
- For Y-STR analysis, victim and reference databases are required.
- For mtDNA analysis, victim and reference databases are required.
- See the [Files](#) section for information on each file.

3. Click the **“Analysis”** button.

The screenshot shows the RelSearch ver. 0.21.0 interface. It has a top navigation bar with 'Load', 'Result', 'Project', 'Database', 'Settings', 'Example files', and 'Manual'. Below this, there are three columns for 'STR', 'Y-STR', and 'mtDNA'. Each column has sections for 'Victim database', 'Reference database', and 'Allele frequencies'. Each section contains a 'Load' button and a file name (e.g., 'str_victim_example.csv', 'y_victim_example.csv', 'mt_victim_example.csv') and an 'Upload complete' button. At the bottom left, there is a blue 'Analyze' button, which is highlighted with an orange arrow.

4. When the analysis is finished, the Result window opens automatically.

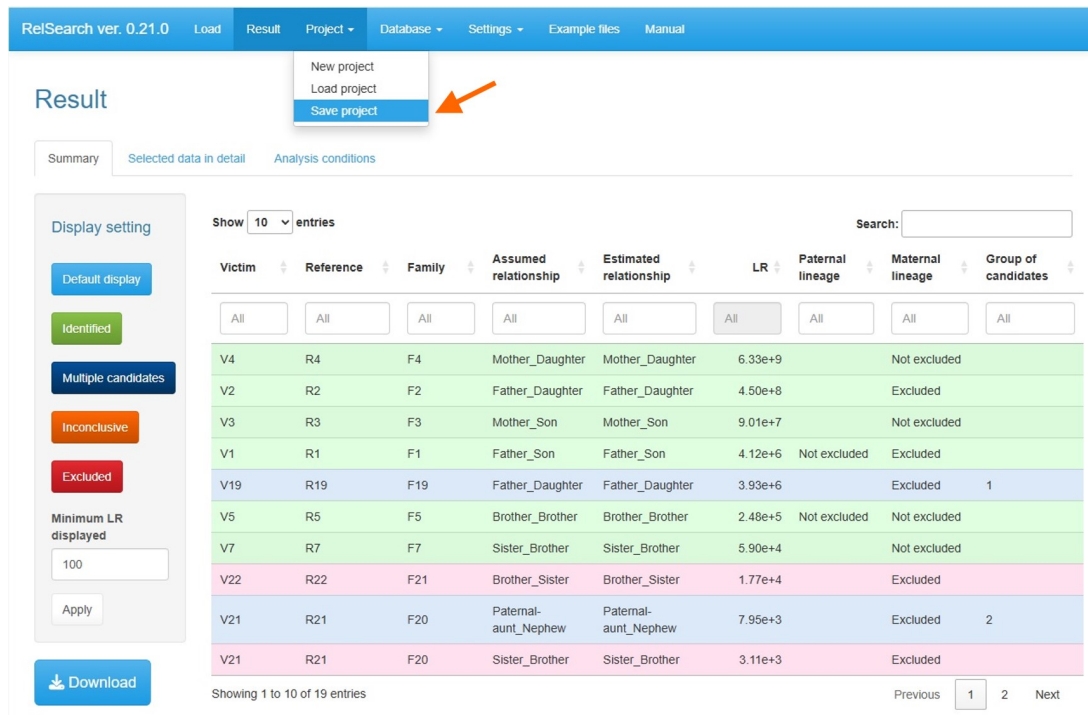
The screenshot shows the RelSearch ver. 0.21.0 Result window. It has a top navigation bar with 'Load', 'Result', 'Project', 'Database', 'Settings', 'Example files', and 'Manual'. Below this, there is a 'Result' section with tabs for 'Summary', 'Selected data in detail', and 'Analysis conditions'. The 'Selected data in detail' tab is active. It shows a table of results with columns: Victim, Reference, Family, Assumed relationship, Estimated relationship, LR, Paternal lineage, Maternal lineage, and Group of candidates. The table contains 19 entries. A 'Display setting' panel on the left allows filtering by 'Identified', 'Multiple candidates', 'Inconclusive', and 'Excluded'. A 'Minimum LR displayed' field is set to 100. A 'Download' button is at the bottom left. A search bar is at the top right. A message box at the top center says 'Data with LR greater than 100 is displayed.'

Victim	Reference	Family	Assumed relationship	Estimated relationship	LR	Paternal lineage	Maternal lineage	Group of candidates
V4	R4	F4	Mother_Daughter	Mother_Daughter	6.33e+9	Not excluded		
V2	R2	F2	Father_Daughter	Father_Daughter	4.50e+8	Excluded		
V3	R3	F3	Mother_Son	Mother_Son	9.01e+7	Not excluded		
V1	R1	F1	Father_Son	Father_Son	4.12e+6	Not excluded	Excluded	
V19	R19	F19	Father_Daughter	Father_Daughter	3.93e+6	Excluded		1
V5	R5	F5	Brother_Brother	Brother_Brother	2.48e+5	Not excluded	Not excluded	
V7	R7	F7	Sister_Brother	Sister_Brother	5.90e+4	Not excluded		
V22	R22	F21	Brother_Sister	Brother_Sister	1.77e+4	Excluded		
V21	R21	F20	Paternal-aunt_Nephew	Paternal-aunt_Nephew	7.95e+3	Excluded		2
V21	R21	F20	Sister_Brother	Sister_Brother	3.11e+3	Excluded		

Note:

- See the [Result](#) section for information on navigating the Result window.

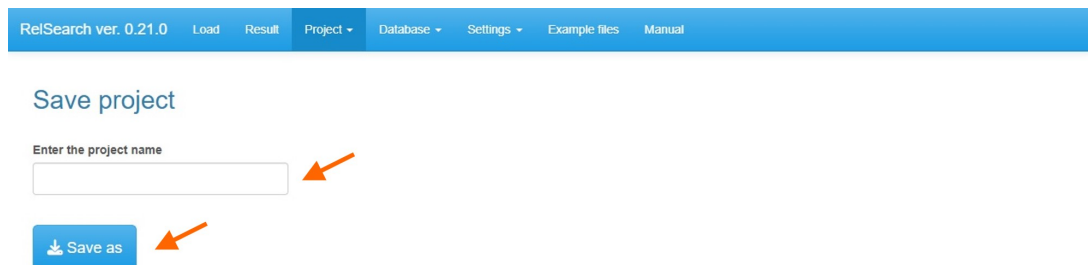
5. Select **Project** ► **Save project**.



The screenshot shows the RelSearch ver. 0.21.0 interface. The 'Project' dropdown menu is open, and the 'Save project' option is highlighted with an orange arrow. The main content area displays a table of results with columns: Victim, Reference, Family, Assumed relationship, Estimated relationship, LR, Paternal lineage, Maternal lineage, and Group of candidates. The table shows 19 entries, with the first 10 displayed. The 'Save project' button is located in the top right corner of the interface.

Victim	Reference	Family	Assumed relationship	Estimated relationship	LR	Paternal lineage	Maternal lineage	Group of candidates
V4	R4	F4	Mother_Daughter	Mother_Daughter	6.33e+9		Not excluded	
V2	R2	F2	Father_Daughter	Father_Daughter	4.50e+8		Excluded	
V3	R3	F3	Mother_Son	Mother_Son	9.01e+7		Not excluded	
V1	R1	F1	Father_Son	Father_Son	4.12e+6	Not excluded	Excluded	
V19	R19	F19	Father_Daughter	Father_Daughter	3.93e+6		Excluded	1
V5	R5	F5	Brother_Brother	Brother_Brother	2.48e+5	Not excluded	Not excluded	
V7	R7	F7	Sister_Brother	Sister_Brother	5.90e+4		Not excluded	
V22	R22	F21	Brother_Sister	Brother_Sister	1.77e+4		Excluded	
V21	R21	F20	Paternal-aunt_Nephew	Paternal-aunt_Nephew	7.95e+3		Excluded	2
V21	R21	F20	Sister_Brother	Sister_Brother	3.11e+3		Excluded	

6. Enter the project name and click the **"Save as"** button.



The screenshot shows the 'Save project' form in the RelSearch interface. The form has a text input field labeled 'Enter the project name' and a 'Save as' button. An orange arrow points to the 'Save as' button. The 'Project' dropdown menu is also visible in the top navigation bar.

Note:

- The saved project can be loaded from **Project** ► **Load project**.
- How to handle projects is described in the [Other functions](#) section.

Chapter 3

Files

3.1 STR: Victim database

SampleName	D3S1358	D3S1358	vWA	vWA	D16S539	D16S539	CSF1PO	CSF1PO
V1	17	17	17		11		12	
V2	17		18	18			10	11
V3	15	17	17	18	10	10	10	12
V4	15	18	15	18	12	12	11	12
V5	15	15	18		9	9	10	

Note:

- File type: .csv
- This file requires the column "SampleName" and columns for each marker (two columns in each).
- The marker with two empty cells (e.g., D16S539 of the sample "V2" in the above table) is ignored when calculating the likelihood ratio.
- The marker with one empty cell (e.g., vWA of the sample "V1" in the above table) can be regarded as both homozygote (i.e., no drop-out) and heterozygote with drop-out of one allele when calculating the likelihood ratio.

3.2 STR: Reference database

SampleName	Family	Relationship	D3S1358	D3S1358	vWA	vWA	D16S539	D16S539
R1	F1	Father_Son	17	17	14	17	10	10
R2	F2	Brother_Brother	16	17	17	17	10	12
R3	F3	Brother_Brother	16	17	14	18	9	9
R3	F3	Father_Son	16	17	14	18	9	9
R4	F4	Brother_Brother	15	16	17		9	12
R5	F4	Uncle_Nephew	15	15	14	16		

Note:

- File type: .csv
- This file requires the columns "SampleName", "Family", "Relationship", and columns for each marker (two columns in each).
- The names of the relationship need to be pre-defined in **Settings ► Relationships** (see section [Setting](#)).
- If there are multiple missing family members for a reference, add rows for each relationship of the members (e.g., Brother_Brother and Father_Son of the sample "R3" in the above table).
- The marker with two empty cells (e.g., D16S539 of the sample "R5" in the above table) is ignored when calculating the likelihood ratio.
- The marker with one empty cell (e.g., vWA of the sample "R4" in the above table) can be regarded as both homozygote (i.e., no drop-out) and heterozygote with drop-out of one allele when calculating the likelihood ratio.

3.3 STR: Allele frequencies

Allele	D3S1358	vWA	D16S539	CSF1PO	TPOX
10			602	648	100
11			562	620	1072
12	6		517	1267	114
13	3	1	209	208	3
14	79	586	26	52	2
15	1192	78	3	14	

Note:

- File type: .csv
- This file requires the columns "Allele" and columns for each marker (one column in each).
- Allele counts in the population database are entered in the columns for each marker.

3.4 Y-STR: Victim database

SampleName	DYS392	DYS518	DYS570	DYS437	DYS385
V1	11	37	17	14	13,17
V2	13	38	19	14	10,20
V3	11	37	17	14	14
V4	11	37	16	14	13
V5			18		13,17

Note:

- File type: .csv
- This file requires the column "SampleName" and columns for each marker (one column in each).
- In the marker with more than one allele, each allele must be separated by a comma without any spaces (e.g., DYS385 in the above table).
- The marker with an empty cell (e.g., DYS392 of the sample "V5" in the above table) is ignored for the analysis.

3.5 Y-STR: Reference database

SampleName	Family	Relationship	DYS392	DYS518	DYS570	DYS437	DYS385
R1	F1	Father_Son	13	38	19	14	10,20
R2	F2	Brother_Brother	11	37	17	14	14,17
R3	F3	Brother_Brother	11	38	16	14	11
R3	F3	Father_Son	11	38	16	14	11
R4	F4	Brother_Brother	11	38	19	14	13,15
R5	F4	Uncle_Nephew	14		18	14	13,18

Note:

- File type: .csv
- This file requires the column "SampleName", "Family", "Relationship", and columns for each marker (one column in each).
- The names of the relationship need to be pre-defined in **Settings ► Relationships** (see section [Setting](#)).
- If there are multiple missing family members for a reference, add rows for each relationship of the members (e.g., Brother_Brother and Father_Son of the sample "R3" in the above table).
- In the marker with more than one allele, each allele must be separated by a comma without any spaces (e.g., DYS385 in the above table).
- The marker with an empty cell (e.g., DYS518 of the sample "R5" in the above table) is ignored for the analysis.

3.6 mtDNA: Victim database

SampleName	Range	Haplotype
V1	73-340 16024-16365	16183C 16189C 73G 263G 309.1C 315.1C
V2	73-340 16024-16365	16223T 16362C 73G 263G 309.1C 315.1C
V3	73-167 240-340 16024-16365	16189C 16223T 16274A 16362C 73G 263G
V4	73-265 16024-16284	16223T 73G 152C 263G
V5		

Note:

- File type: .csv
- This file requires the column "SampleName", "Range", and "Haplotype".
- In the "Range" column, the first and the last positions in each readable sequence must be written with a hyphen between these positions (e.g, 73-340). The readable sequences must be separated by a single blank space.
- In the "Haplotype" column, the mtDNA types are expressed as the differences to a reference sequence such as the revised Cambridge Reference Sequence [1] (e.g., 73G, 315.1C, and so on). Each mtDNA type must be separated by a single blank space.
- The empty cells in the "Range" and "Haplotype" columns (e.g., the sample "V5" in the above table) mean that there is no readable sequence.

3.7 mtDNA: Reference database

SampleName	Family	Relationship	Range	Haplotype
R1	F1	Mother_Daughter	73-340 16024-16365	16223T 73G 263G 309.1C 315.1C
R2	F2	Mother_Daughter	73-340 16024-16365	16172C 16223T 73G 150T 263G
R3	F3	Brother_Brother	73-340 16024-16365	16172C 16189C 16223T 73G 263G
R3	F3	Father_Son	73-265 16024-16284	16172C 16189C 16223T 73G 263G
R4	F4	Sister_Sister	73-340 16024-16365	16223T 16362C 73G 263G 315.1C
R5	F4	Aunt_Niece		

Note:

- File type: .csv
- This file requires the column "SampleName", "Family", "Relationship", "Range", and "Haplotype".
- The names of the relationship need to be pre-defined in **Settings ► Relationships** (see section [Setting](#)).
- If there are multiple missing family members for a reference, add rows for each relationship of the members (e.g., Brother_Brother and Father_Son of the sample "R3" in the above table).
- In the "Range" column, the first and the last positions in each readable sequence must be written with a hyphen between these positions (e.g, 73-340). The readable sequences must be separated by a single blank space.
- In the "Haplotype" column, the mtDNA types are expressed as the differences to a reference sequence such as the revised Cambridge Reference Sequence [\[1\]](#) (e.g., 73G, 315.1C, and so on). Each mtDNA type must be separated by a single blank space.
- The empty cells in the "Range" and "Haplotype" columns (e.g., the sample "R5" in the above table) mean that there is no readable sequence.

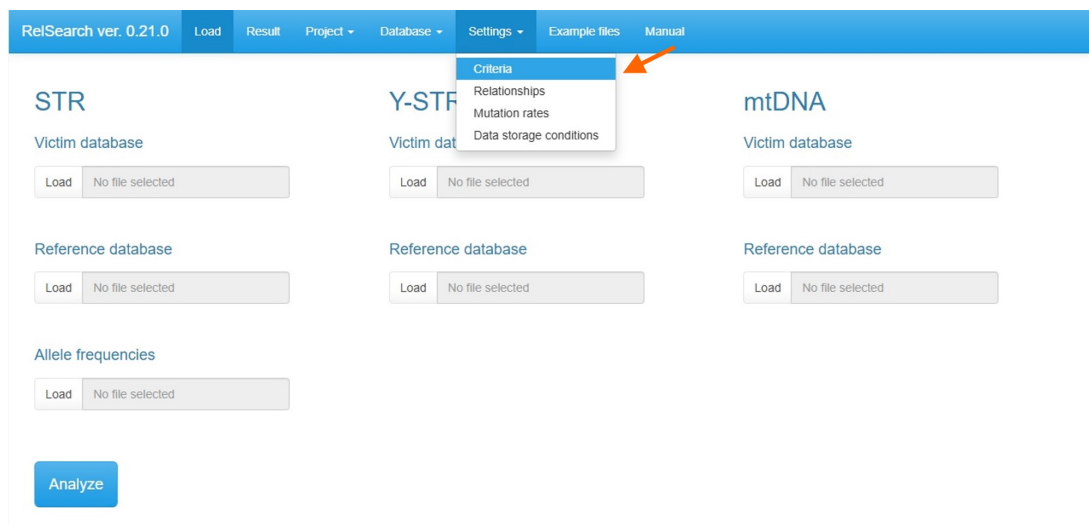
Chapter 4

Setting

4.1 Criteria

In RelSearch, users can set some criteria to identify or exclude the assumed relationship for each victim-reference pair. For autosomal STR, RelSearch calculates LR values for each victim-reference pair and investigates whether these LR values are greater than an LR threshold or not. For Y-STR, RelSearch analyzes the number of mismatched loci and the total mismatched steps between victim and reference Y-STR profiles to search paternal lineages. For mtDNA, RelSearch investigates the number of mismatched nucleotides between the victim and the reference sequences to search maternal lineages.

To set criteria, select **Setting ► Criteria**.



Then the criteria setting window will open automatically.

RelSearch ver. 0.21.0 Load Result Project Database Settings Example files Manual

LR Criteria

Minimum LR to support the assumed relationship

Maximum LR to exclude the assumed relationship

Y-STR Usage Criteria

Minimum number of loci with genotypes

Paternal Lineage Criteria

Maximum number of mismatched loci

Upper limit of the total mutational steps

mtDNA Usage Criteria

Minimum read length (bp)

Maternal Lineage Criteria

Maximum number of inconsistency

4.1.1 LR Criteria

- Minimum LR to support the assumed relationship: If the LR is greater than the set value, the victim-reference pair is supported as the assumed relationship.
- Maximum LR to exclude the assumed relationship: If the LR is less than the set value, the victim-reference pair is excluded from the assumed relationship.

Note:

If the LR is between the above two values, the relationship of the victim-reference pair is considered inconclusive.

4.1.2 Y-STR Usage Criteria

- Minimum number of loci with genotypes: To adopt the Y-STR analysis result of a victim-reference pair, the number of loci where genotypes of both individuals are obtained must be greater than the set value. If the number is less than the set value, the paternal lineage of the victim-reference pair is considered inconclusive.

Note:

The victim-reference pair that does not meet the following Paternal Lineage Criteria is excluded from the paternal lineage even if the pair does not meet the Y-STR Usage Criteria.

4.1.3 Paternal Lineage Criteria

- Maximum number of mismatched loci: If the number of mismatched loci in a victim-reference pair is greater than the set value, the pair is excluded from the paternal lineage.
- Upper limit of the total mutational steps: If the total mutational steps in a victim-reference pair is greater than the set value, the pair is excluded from the paternal lineage.

4.1.4 mtDNA Usage Criteria

- Minimum read length (bp): To adopt the mtDNA analysis result of a victim-reference pair, the read length in both individuals must be greater than the set value. If the number is less than the set value, the maternal lineage of the victim-reference pair is considered inconclusive.

Note:

The victim-reference pair that does not meet the following Maternal Lineage Criteria is excluded from the maternal lineage even if the pair does not meet the mtDNA Usage Criteria.

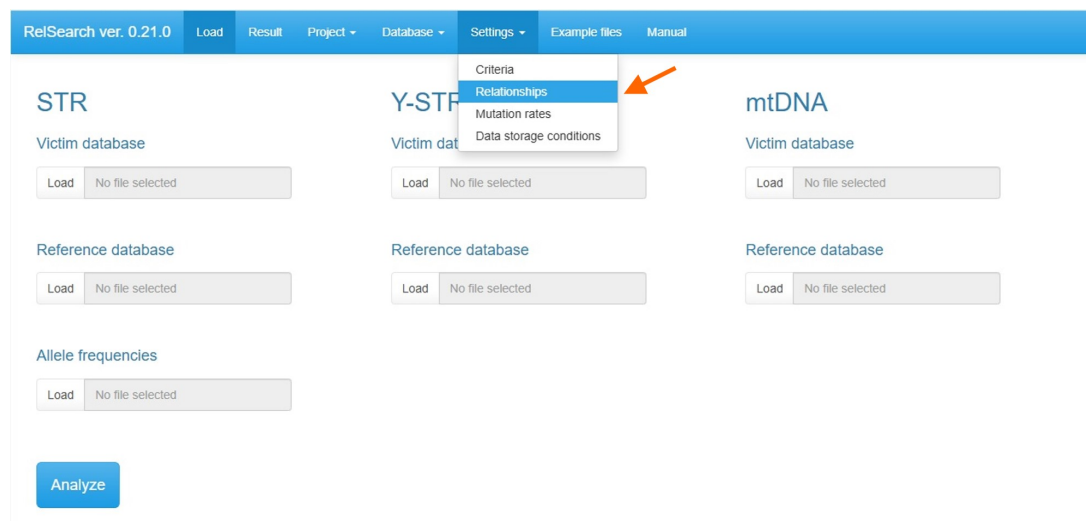
4.1.5 Maternal Lineage Criteria

- Maximum number of inconsistency: If the number of inconsistency in a victim-reference pair is greater than the set value, the pair is excluded from the maternal lineage.

After changing the criteria, press the **Save** button to reflect the changes. Press the **Reset** button if you want to return to default values.

4.2 Relationship

The names of the relationship in each reference profile need to be pre-defined. To define the relationship names, select **Setting ► Relationships**.



Then the relationship setting window will open automatically.

RelSearch ver. 0.21.0 Load Result Project ▾ Database ▾ Settings ▾ Example files Manual

Relationships

Edit
Add
Delete
Reset
Family tree

Show 10 entries Search:

Relationship	Sex (Victim)	Sex (Reference)	Pr (IBD = 2)	Pr (IBD = 1)	Pr (IBD = 0)	Paternal lineage	Maternal lineage
Father_Son	M	M	0	1	0	Yes	No
Father_Daughter	M	F	0	1	0	No	No
Mother_Son	F	M	0	1	0	No	Yes
Mother_Daughter	F	F	0	1	0	No	Yes
Son_Father	M	M	0	1	0	Yes	No
Son_Mother	M	F	0	1	0	No	Yes
Daughter_Father	F	M	0	1	0	No	No
Daughter_Mother	F	F	0	1	0	No	Yes
Brother_Brother	M	M	0.25	0.5	0.25	Yes	Yes
Brother_Sister	M	F	0.25	0.5	0.25	No	Yes

Showing 1 to 10 of 44 entries

Previous
1
2
3
4
5
Next

There are 44 pre-defined victim-reference relationships as the default. The names of these relationships are determined in the order of victim_reference (e.g., Father_Son means that a victim is the father and a reference is the son). You are free to change these names.

Information on the table is as follows.

- **Sex (Victim):** The biological sex of the victim
- **Sex (Reference):** The biological sex of the reference
- **Pr(IBD = 2):** The probability that two alleles are identity by descent (IBD).
- **Pr(IBD = 1):** The probability that one allele is identity by descent (IBD).
- **Pr(IBD = 0):** The probability that zero allele is identity by descent (IBD).
- **Paternal lineage:** Information on whether each relationship is paternal lineage or not.
- **Maternal lineage:** Information on whether each relationship is maternal lineage or not.

You can edit, add, and delete the relationships from the left sidebar.

4.2.1 Edit

You can edit the relationship names from the **Edit** button.

RelSearch ver. 0.21.0

Relationship

Edit

Add

Delete

Reset

Family tree

Maternal lineage

No

No

Yes

Yes

Yes

Son_Father M M 0 1 0 Yes No

Son_Mother M F 0 1 0 No Yes

Daughter_Father F M 0 1 0 No No

Daughter_Mother F F 0 1 0 No Yes

Brother_Brother M M 0.25 0.5 0.25 Yes Yes

Brother_Sister M F 0.25 0.5 0.25 No Yes

Showing 1 to 10 of 44 entries

Previous 1 2 3 4 5 Next

Edit the name of the relationship

Select a relationship

Father_Son

Enter a new name

Save Cancel

In the pop-up window, select a relationship from the defined relationships, and enter a new name of the selected relationship. Press the **Save** button to reflect the new name.

4.2.2 Add

You can define a new relationship from the **Add** button. Then, the following pop-up window will open.

Add information on a relationship

Relationship

Family tree

Incorrect setting of the family tree!

Set a family tree

Add a person

Delete a person

View family tree

Person	Sex	Father	Mother	Founder
Victim	M	Victim	Victim	No
Ref	M	Victim	Victim	No

Save

Cancel

Enter a relationship name, and make a family tree to define the relationship between a victim and a reference. The following is a list of functions to make a family tree.

- **Add a person:** The function to add an unknown person (UK) to the list of persons.
- **Delete a person:** The function to delete an unknown person (UK) from the list of person.
- **Sex:** The function to select a biological sex (male: M or female: F).
- **Father:** The function to select a person who is the father.
- **Mother:** The function to select a person who is the mother.
- **Founder:** The function to select whether the person is a founder of the family tree or not.

Here is an example of defining the brother-brother relationship for the victim-reference pair.

1. Press the **Add** button twice to add the untyped father and mother. UK1 and UK2 are assumed to be the father and mother, respectively.
2. Set the sex of the Victim, Ref, UK1, and UK2 to "M", "M", "M", and "F", respectively.
3. In the Founder column, select "No" for Victim and Ref, and select "Yes" for UK1 and UK2.
4. In the Father column, select "UK1" for Victim and Ref.
5. In the Mother column, select "UK2" for Victim and Ref.

Set a family tree

Person	Sex	Father	Mother	Founder
Victim	M	UK1	UK2	No
Ref	M	UK1	UK2	No
UK1	M	Victim	Victim	Yes
UK2	F	Victim	Victim	Yes

Press the **View family tree** button to check whether the relationship between a victim and a reference is correctly defined or not.

[Add information on a relationship](#)

Relationship

Family tree

Incorrect setting of the family tree!

Set a family tree

Add a person

Delete a person

View family tree

Person	Sex	Father	Mother	Founder
Victim	M	UK1	UK2	No
Ref	M	UK1	UK2	No
UK1	M	Victim	Victim	Yes
UK2	F	Victim	Victim	Yes

Save

Cancel

If the relationship between a victim and a reference is correctly defined, the family tree will be displayed.

Add information on a relationship

Relationship

Family tree

Set a family tree

Add a person
Delete a person
View family tree

Person	Sex	Father	Mother	Founder
Victim	M	UK1	UK2	No
Ref	M	UK1	UK2	No
UK1	M	Victim	Victim	Yes
UK2	F	Victim	Victim	Yes

Save
Cancel

Press the **Save** button.

Add information on a relationship

Relationship

Test

Family tree

```
graph TD; UK1[UK1] --- UK2[UK2]; UK1 --- Ref[Ref]; UK2 --- Victim[Victim];
```

Set a family tree

Add a person Delete a person View family tree

Person	Sex	Father	Mother	Founder
Victim	M	UK1	UK2	No
Ref	M	UK1	UK2	No
UK1	M	Victim	Victim	Yes
UK2	F	Victim	Victim	Yes

Save Cancel

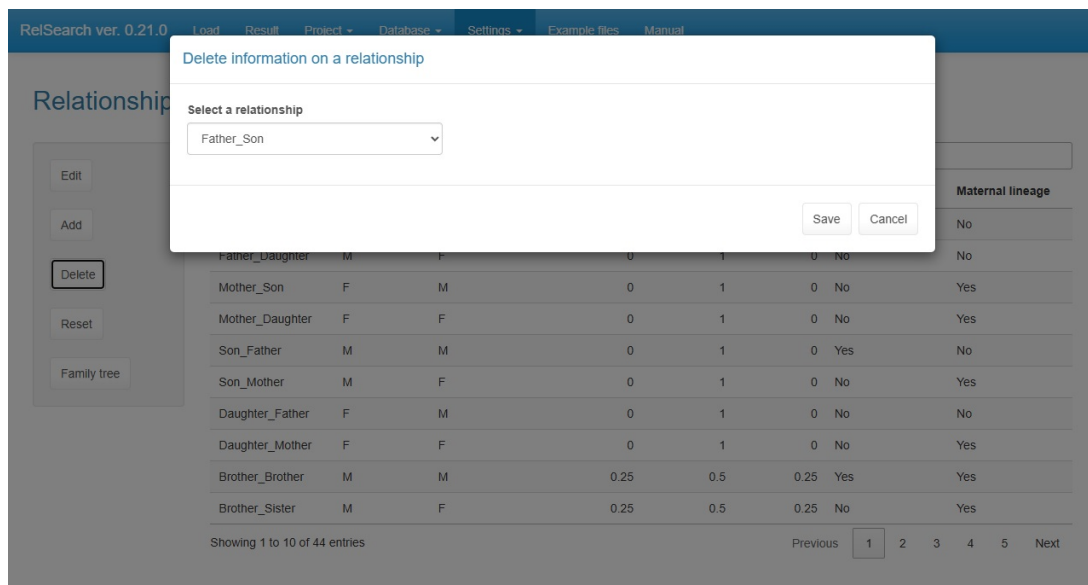
Note:

Information on the biological sex, IBD probabilities, paternal lineage, and maternal lineage in the new relationship is automatically determined based on your-defined family tree.

4.2.3 Delete

You can delete a defined relationship from the **Delete** button.

In the pop-up window, select a relationship from the defined relationships. Press the **Save** button to delete the selected relationship.



4.2.4 Reset

If you want to return to default settings, press the **Reset** button. In the pop-up window, press the **Restore default** button.

The screenshot shows the RelSearch application interface. A modal dialog box titled "Reset information on the relationship" is open, displaying the text "Click Restore default to remove all changes." and two buttons: "Restore default" and "Cancel".

The background interface includes a top navigation bar with the following items: RelSearch ver. 0.21.0, Load, Result, Project, Database, Settings, Example files, and Manual. The main content area is titled "Relationships" and features a sidebar with buttons: Edit, Add, Delete, Reset (highlighted), and Family tree. A "Show 10" dropdown is visible. The main table lists various relationships with columns for relationship type, gender, and lineage status.

Relationship	Gender	Paternal lineage	Maternal lineage
Father_Son	M	0	1
Father_Daughter	M	0	1
Mother_Son	F	0	1
Mother_Daughter	F	0	1
Son_Father	M	0	1
Son_Mother	M	0	1
Daughter_Father	F	0	1
Daughter_Mother	F	0	1
Brother_Brother	M	0.25	0.5
Brother_Sister	M	0.25	0.5

At the bottom, it indicates "Showing 1 to 10 of 44 entries" and a pagination control with "Previous", "1" (selected), "2", "3", "4", "5", and "Next".

4.2.5 Family tree

You can check the family tree of each defined relationship. Select a row from the relationship table and press the **Family tree** button.

RelSearch ver. 0.21.0 Load Result Project Database Settings Example files Manual

Relationships

Show 10 entries Search:

Relationship	Sex (Victim)	Sex (Reference)	Pr (IBD = 2)	Pr (IBD = 1)	Pr (IBD = 0)	Paternal lineage	Maternal lineage
Father_Son	M	M	0	1	0	Yes	No
Father_Daughter	M	F	0	1	0	No	No
Mother_Son	F	M	0	1	0	No	Yes
Mother_Daughter	F	F	0	1	0	No	Yes
Son_Father	M	M	0	1	0	Yes	No
Son_Mother	M	F	0	1	0	No	Yes
Daughter_Father	F	M	0	1	0	No	No
Daughter_Mother	F	F	0	1	0	No	Yes
Brother_Brother	M	M	0.25	0.5	0.25	Yes	Yes
Brother_Sister	M	F	0.25	0.5	0.25	No	Yes

Showing 1 to 10 of 44 entries Previous 1 2 3 4 5 Next

Buttons: Edit, Add, Delete, Reset, Family tree

Then, the family tree will be displayed.

RelSearch ver. 0.21.0 Load Result Project Database Settings Example files Manual

Relationships

Show 10 entries Search:

Relationship	Sex (Victim)	Sex (Reference)	Pr (IBD = 2)	Pr (IBD = 1)	Pr (IBD = 0)	Paternal lineage	Maternal lineage
Father_Son	M	M	0	1	0	Yes	No
Father_Daughter	M	F	0	1	0	No	No
Mother_Son	F	M	0	1	0	No	Yes
Mother_Daughter	F	F	0	1	0	No	Yes
Son_Father	M	M	0	1	0	Yes	No
Son_Mother	M	F	0	1	0	No	Yes
Daughter_Father	F	M	0	1	0	No	No
Daughter_Mother	F	F	0	1	0	No	Yes
Brother_Brother	M	M	0.25	0.5	0.25	Yes	Yes
Brother_Sister	M	F	0.25	0.5	0.25	No	Yes

Showing 1 to 10 of 44 entries Previous 1 2 3 4 5 Next

Buttons: Edit, Add, Delete, Reset, Family tree

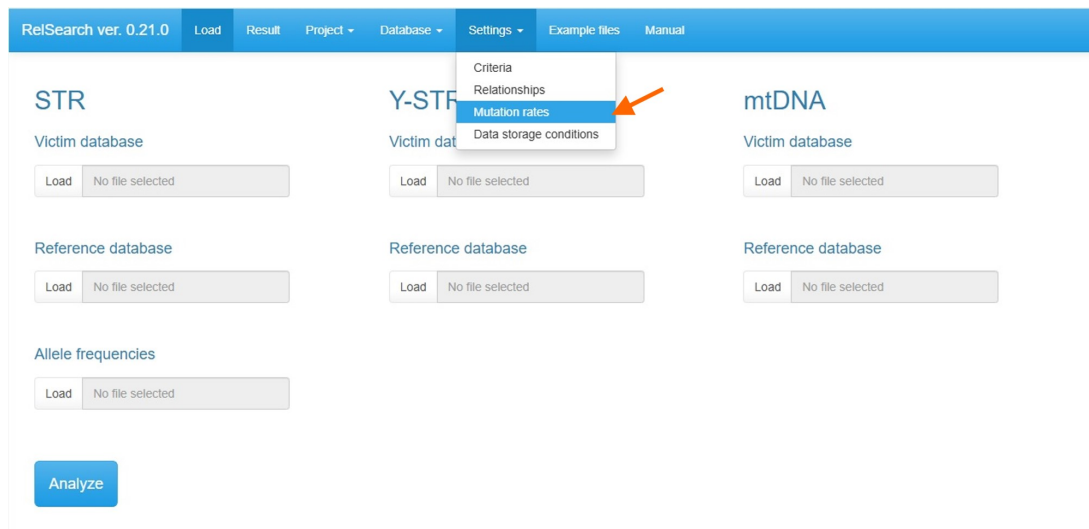
Relationship: Father_Son

Close

4.3 Mutation rate

Mutation rates of each autosomal STR marker need to be set to calculate the likelihood ratio (LR) for parent-child relationship. The default mutation rates considering the difference between paternal and maternal and mutational steps (-1, +1, -2, and +2) are reported by Morimoto et al [2].

To set mutation rates, select **Setting ► Mutation rates**.



Then the setting window for mutation rates will open automatically.

The screenshot shows the 'Mutation rates' setting window. On the left is a control panel with 'Edit', 'Add', 'Delete', and 'Reset' buttons. The main area displays a table with 11 columns: Locus, Paternal -2 step, Paternal -1 step, Paternal 0 step, Paternal +1 step, Paternal +2 step, Maternal -2 step, Maternal -1 step, Maternal 0 step, Maternal +1 step, and Maternal +2 step. The table lists 20 STR markers with their corresponding mutation rates. A search bar is located at the top right of the table area.

Locus	Paternal -2 step	Paternal -1 step	Paternal 0 step	Paternal +1 step	Paternal +2 step	Maternal -2 step	Maternal -1 step	Maternal 0 step	Maternal +1 step	Maternal +2 step
D3S1338	0.000037	0.001423565	0.997353441	0.001148329	0.0000377	0	0.0001927	0.99969733	0.0000756	0.0000344
VWA	0.000146413	0.002718279	0.994905627	0.002200627	0.0000292	0	0.000125004	0.999377764	0.000444731	0.0000525
D16S539	0.0000841	0.00093427	0.997628583	0.001353042	0	0	0.000484253	0.999411835	0.000103912	0
CSF1PO	0	0.002042983	0.995111085	0.001845931	0	0	0.000212751	0.99940775	0.0003795	0
TPOX	0	0.000115566	0.999697045	0.000187388	0	0	0.0000751	0.999848864	0.0000761	0
D8S1179	0.0000482	0.001301255	0.997430135	0.001196466	0.0000239	0	0.000133755	0.999702176	0.000141848	0.0000222
D21S11	0.000037	0.00067972	0.998012967	0.001082339	0	0.000036	0.000107823	0.999727236	0.000128943	0
D18S51	0.00014735	0.000942863	0.997459246	0.001420767	0.0000298	0.0000213	0.000117496	0.999364128	0.000475601	0.0000215
D2S441	0.0000544	0.001166981	0.997463648	0.001250749	0.0000642	0.0000149	0.000270098	0.999494265	0.000212047	0.0000871
D19S433	0	0.000958424	0.998425905	0.000452336	0.000164235	0	0.000004888	0.99943539	0.0000597	0
TH01	0	0.0000448	0.999895495	0.0000597	0	0.000017	0.0000546	0.999920126	0.00000829	0
FGA	0.000277755	0.001497345	0.995437569	0.002707964	0.0000794	0	0.000348321	0.999358247	0.000293432	0
D22S1045	0.0000544	0.001166981	0.997463648	0.001250749	0.0000642	0.0000149	0.000270098	0.999494265	0.000212047	0.0000871
D5S818	0	0.001420549	0.99658405	0.001198	0.000540952	0	0.000258823	0.999462401	0.000276776	0
D13S317	0	0.001369203	0.997193828	0.001397392	0.0000396	0.0000333	0.0000708	0.999657082	0.000238821	0
D7S820	0	0.001302502	0.997827141	0.000870357	0	0	0.0000622	0.999813227	0.000124612	0
SE33	0.0000544	0.001166981	0.997463648	0.001250749	0.0000642	0.0000149	0.000270098	0.999494265	0.000212047	0.0000871
D10S1248	0.0000544	0.001166981	0.997463648	0.001250749	0.0000642	0.0000149	0.000270098	0.999494265	0.000212047	0.0000871
D15S656	0.0000544	0.001166981	0.997463648	0.001250749	0.0000642	0.0000149	0.000270098	0.999494265	0.000212047	0.0000871

4.3.1 Edit

You can edit the mutation rates from the **Edit** button.

In the pop-up window, select a locus and change the mutation rates. Press the **Save** button to reflect the changes.

RelSearch ver. 0.21.0

Load Result Project Database Settings Example files Manual

Mutation rate

Edit

Add

Delete

Reset

Edit a mutation rate

Select a locus

D3S1358

Paternal -2 step

0.000037

Paternal -1 step

0.001423565

Paternal +1 step

0.001148329

Paternal +2 step

0.0000377

Maternal -2 step

0

Maternal -1 step

0.0001927

Maternal +1 step

0.0000756

Maternal +2 step

0.0000344

Save Cancel

D19S433	0	0.000958424	0.998425005	0.000452336	0.000164235	0	0.000504888	0.99943539	0.0000597
TH01	0	0.0000448	0.999895495	0.0000597	0	0.000017	0.0000546	0.999920126	0.00000829
FGA	0.000277755	0.001497345	0.995437569	0.002707964	0.0000794	0	0.000348321	0.999358247	0.000293432
D22S1045	0.0000544	0.001166981	0.997463648	0.001250749	0.0000642	0.0000149	0.000270098	0.999494265	0.000212047

4.3.2 Add

You can add a locus from the **Add** button.

In the pop-up window, enter a locus name and enter the mutation rates. Press the **Save** button to add the locus.

Add a mutation rate

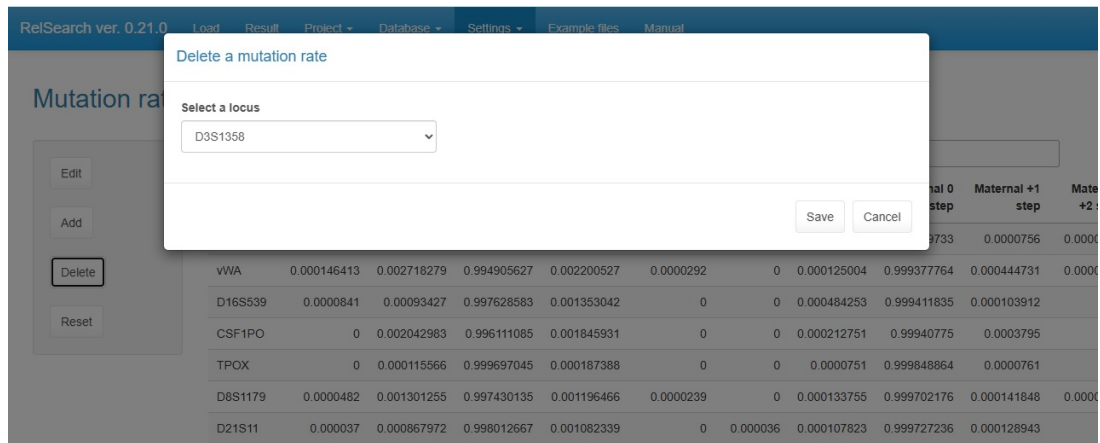
Locus name 	
Paternal -2 step 	Maternal -2 step
Paternal -1 step 	Maternal -1 step
Paternal +1 step 	Maternal +1 step
Paternal +2 step 	Maternal +2 step

Save Cancel

4.3.3 Delete

You can delete a locus from the **Delete** button.

In the pop-up window, select a locus. Press the **Save** button to delete the selected locus.



4.3.4 Reset

If you want to return to default settings, press the **Reset** button. In the pop-up window, press the **Restore default** button.

The screenshot shows the RelSearch ver. 0.21.0 interface. The 'Mutation rates' section is active, displaying a table of mutation rates for various loci. A 'Reset' button is visible in the left sidebar. A dialog box titled 'Reset the mutation rates' is open, prompting the user to click 'Restore default' to remove all changes. The dialog box also includes a 'Cancel' button.

RelSearch ver. 0.21.0 Load Result Project Database Settings Example files Manual

Mutation rates

Show 50

Edit

Add

Delete

Reset

Search:

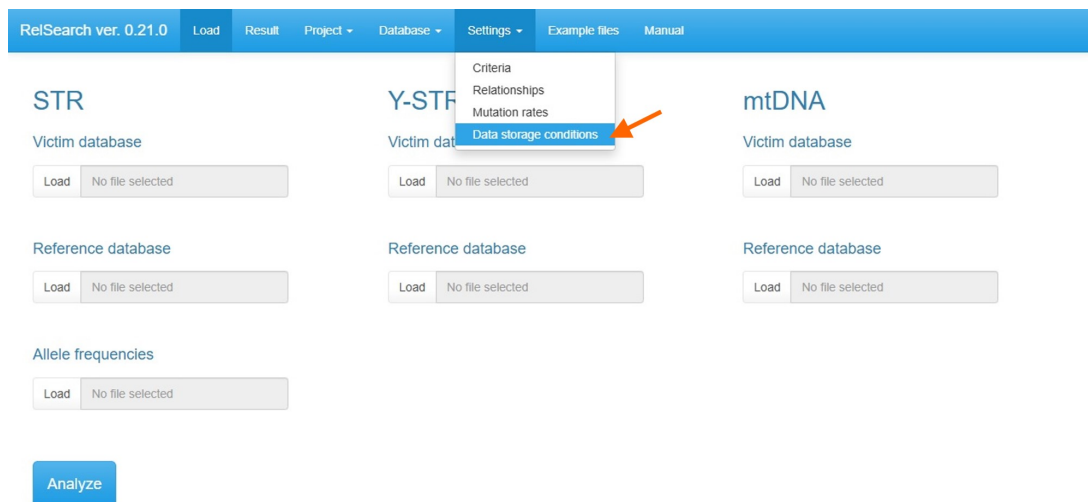
Maternal -1 step Maternal 0 step Maternal +1 step Maternal +2 step

Locus	step	step	step	step	step	-2 step	Maternal -1 step	Maternal 0 step	Maternal +1 step	Maternal +2 step
D3S1358	0.000037	0.001423565	0.997353441	0.001148329	0.0000377	0	0.0001927	0.99969733	0.0000756	0.0000377
vWA	0.000146413	0.002718279	0.994905627	0.002200527	0.0000292	0	0.000125004	0.999377764	0.000444731	0.0000292
D16S539	0.0000841	0.00093427	0.997628583	0.001353042	0	0	0.000484253	0.999411835	0.000103912	0.0000841
CSF1PO	0	0.002042983	0.996111085	0.001845931	0	0	0.000212751	0.99940775	0.0003795	0
TPOX	0	0.000115566	0.999697045	0.000187388	0	0	0.0000751	0.999848864	0.0000761	0
D8S1179	0.0000482	0.001301255	0.997430135	0.001196466	0.0000239	0	0.000133755	0.999702176	0.000141848	0.0000239
D21S11	0.000037	0.000867972	0.998012667	0.001082339	0	0.000036	0.000107823	0.999727236	0.000128943	0.000036

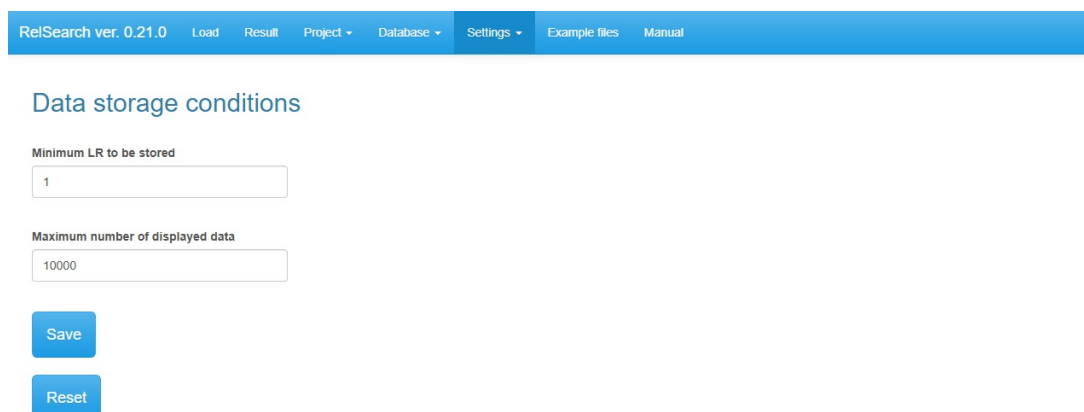
4.4 Data storage conditions

When handling a large number of victim and reference profiles, it may be difficult to store all the results because RelSearch may run out of memory. You can set the data storage conditions.

To set the conditions, select **Setting ► Data storage conditions**.



Then the setting window for data storage will open automatically.



The screenshot shows the 'Settings' window of RelSearch ver. 0.21.0. The window has a blue header bar with the version and menu items: Load, Result, Project, Database, Settings (active), Example files, and Manual. Below the header, the title 'Data storage conditions' is displayed. There are two input fields: 'Minimum LR to be stored' with the value '1' and 'Maximum number of displayed data' with the value '10000'. At the bottom, there are two blue buttons: 'Save' and 'Reset'.

- Minimum LR to be stored: The minimum LR to be stored in the project data. If you want to reduce the file size of the project data, the minimum LR should be increased.
- Maximum number of displayed data: The maximum number of data displayed in the Results tab. If this number is set too high, RelSearch may run out of memory.

Press the **Save** button to reflect the changes. Press the **Reset** button if you want to return to default values.

Chapter 5

Result

This section describes the display of results on the Results tab.

5.1 Summary

When the analysis is complete, summary data is automatically displayed. You can change the data displayed using the buttons in the left sidebar. You can also download the displayed data as a .csv file using the **Download** button.

ReSearch ver. 0.21.0 Load Result Project Database Settings Example files Manual

Result

Summary Selected data in detail Analysis conditions

Display setting

Default display

Identified

Multiple candidates

Inconclusive

Excluded

Minimum LR displayed

100

Apply

Download

Show 10 entries

Search:

Victim	Reference	Family	Assumed relationship	Estimated relationship	LR	Paternal lineage	Maternal lineage	Group of candidates
All	All	All	All	All	All	All	All	All
V4	R4	F4	Mother_Daughter	Mother_Daughter	6.33e+9		Not excluded	
V2	R2	F2	Father_Daughter	Father_Daughter	4.50e+8		Excluded	
V3	R3	F3	Mother_Son	Mother_Son	9.01e+7		Not excluded	
V1	R1	F1	Father_Son	Father_Son	4.12e+6	Not excluded	Excluded	
V19	R19	F19	Father_Daughter	Father_Daughter	3.93e+6		Excluded	1
V5	R5	F5	Brother_Brother	Brother_Brother	2.48e+5	Not excluded	Not excluded	
V7	R7	F7	Sister_Brother	Sister_Brother	5.90e+4		Not excluded	
V22	R22	F21	Brother_Sister	Brother_Sister	1.77e+4		Excluded	
V21	R21	F20	Paternal-aunt_Nephew	Paternal-aunt_Nephew	7.95e+3		Excluded	2
V21	R21	F20	Sister_Brother	Sister_Brother	3.11e+3		Excluded	

Showing 1 to 10 of 19 entries

Previous 1 2 Next

Default display

Default display depends on whether the input data includes autosomal STR data or not.

- If the input data includes autosomal STR data, the data that satisfies the criterion of the 'Minimum LR to support the assumed relationship' (see section Criteria) is displayed.
- If the input data is composed of only Y-STR and/or mtDNA data, data that is not excluded from paternal or maternal lineages (see section Criteria) is displayed.
- If the number of data that satisfies the above conditions is more than the upper limit (i.e., the maximum number of displayed data explained in section [Data storage conditions](#)), the number of data displayed is the upper limit.

Identified

If the data supports that a victim (or a reference) has the assumed relationship with only one reference (or victim), the victim-reference pair is classified as "Identified". The LR must be greater than 'Minimum LR to support the assumed relationship' (see section Criteria) for classification as "Identified". In addition, if the assumed relationship is paternal or maternal lineage, the paternal or maternal lineage criteria (see section Criteria) must be also satisfied.

Multiple candidates

If the data supports that a victim (or a reference) has the assumed relationship with multiple references (or victims), the victim-reference pair is classified as "Multiple candidates". The LR must be greater than 'Minimum LR to support the assumed relationship' (see section Criteria) for classification as "Multiple candidates". In addition, if the assumed relationship is paternal or maternal lineage, the paternal or maternal lineage criteria (see section Criteria) must be also satisfied.

Inconclusive

The victim-reference pair is classified as "Inconclusive" if the pair is not classified in any of the other categories. The "Inconclusive" category includes the following data.

- The LR is between 'Maximum LR to exclude the assumed relationship' and 'Minimum LR to support the assumed relationship' (see section Criteria).
- The data does not meet the Y-STR Usage Criteria when the victim-reference pair is not excluded from the paternal lineage (see section Criteria).
- The data does not meet the mtDNA Usage Criteria when the victim-reference pair is not excluded from the maternal lineage (see section Criteria).
- Autosomal STR data is not available.

Excluded

The victim-reference pair is excluded from the assumed relationship according to the following conditions:

- The LR is less than 'Maximum LR to exclude the assumed relationship' (see section Criteria).
- The paternal or maternal lineage criteria (see section Criteria) is not satisfied when the assumed relationship is paternal or maternal lineage.

Minimum LR displayed

You can change the minimum LR displayed. After setting the value, press the **Apply** button. Please note that the amount of data stored in the project data is limited according to the data storage conditions (see [Data storage conditions](#)). For example, even if you set the minimum LR displayed to less than 1, you cannot check the data with $LR < 1$ as long as the minimum LR to be stored in the project data is set to 1.

5.2 Selected data in detail

The detailed data of a selected victim-reference pair is displayed in **Selected data in detail** tab.

1. Select a row in the summary table.

ReSearch ver. 0.21.0 Load Result Project Database Settings Example files Manual

Result

Summary Selected data in detail Analysis conditions

Display setting

Default display

Identified

Multiple candidates

Inconclusive

Excluded

Minimum LR displayed

100

Apply

Download

Show 10 entries

Search:

Victim	Reference	Family	Assumed relationship	Estimated relationship	LR	Paternal lineage	Maternal lineage	Group of candidates
All	All	All	All	All	All	All	All	All
V4	R4	F4	Mother_Daughter	Mother_Daughter	6.33e+9		Not excluded	
V2	R2	F2	Father_Daughter	Father_Daughter	4.50e+8		Excluded	
V3	R3	F3	Mother_Son	Mother_Son	9.01e+7		Not excluded	
V1	R1	F1	Father_Son	Father_Son	4.12e+6	Not excluded	Excluded	
V19	R19	F19	Father_Daughter	Father_Daughter	3.93e+6		Excluded	1
V5	R5	F5	Brother_Brother	Brother_Brother	2.46e+5	Not excluded	Not excluded	
V7	R7	F7	Sister_Brother	Sister_Brother	5.90e+4		Not excluded	
V22	R22	F21	Brother_Sister	Brother_Sister	1.77e+4		Excluded	
V21	R21	F20	Paternal-aunt_Nephew	Paternal-aunt_Nephew	7.95e+3		Excluded	2
V21	R21	F20	Sister_Brother	Sister_Brother	3.11e+3		Excluded	

Showing 1 to 10 of 19 entries

Previous 1 2 Next

2. Go to **Selected data in detail** tab.

ReSearch ver. 0.21.0 Load Result Project Database Settings Example files Manual

Result

Summary Selected data in detail Analysis conditions

Display setting

Default display

Identified

Multiple candidates

Inconclusive

Excluded

Minimum LR displayed

100

Apply

Download

Show 10 entries

Search:

Victim	Reference	Family	Assumed relationship	Estimated relationship	LR	Paternal lineage	Maternal lineage	Group of candidates
All	All	All	All	All	All	All	All	All
V4	R4	F4	Mother_Daughter	Mother_Daughter	6.33e+9		Not excluded	
V2	R2	F2	Father_Daughter	Father_Daughter	4.50e+8		Excluded	
V3	R3	F3	Mother_Son	Mother_Son	9.01e+7		Not excluded	
V1	R1	F1	Father_Son	Father_Son	4.12e+6	Not excluded	Excluded	
V19	R19	F19	Father_Daughter	Father_Daughter	3.93e+6		Excluded	1
V5	R5	F5	Brother_Brother	Brother_Brother	2.46e+5	Not excluded	Not excluded	
V7	R7	F7	Sister_Brother	Sister_Brother	5.90e+4		Not excluded	
V22	R22	F21	Brother_Sister	Brother_Sister	1.77e+4		Excluded	
V21	R21	F20	Paternal-aunt_Nephew	Paternal-aunt_Nephew	7.95e+3		Excluded	2
V21	R21	F20	Sister_Brother	Sister_Brother	3.11e+3		Excluded	

Showing 1 to 10 of 19 entries

Previous 1 2 Next

In **Selected data in detail** tab, you can review and download the analyzed STR, Y-STR, and mtDNA data for the selected victim-reference pair. You can check all members of the family to which the selected reference belongs.

STR tab

Summary

Selected data in detail

Analysis conditions

Selected victim

VS

Selected reference

RS

Estimated relationship

Brother_Brother

Paternal lineage

Not excluded

Maternal lineage

Not excluded

All family members

RS

Download (STR)

Download (Y-STR)

Download (mtDNA)

STR	Y-STR	mtDNA			
Locus	Victim profile	Reference profile	Likelihood (related)	Likelihood (unrelated)	LR
D3S1358	15, 17	15, 17	0.0704	0.0257	2.74
VWA	16, 18	14, 19	0.000584	0.00234	0.250
D16S839	9, 12	9, 13	0.00369	0.00615	0.599
CSF1PO	10, 10	12, 12	0.00206	0.00823	0.250
TPOX	11, 11	10, 11	0.00290	0.00304	0.951
D8S1179	13, 13	11, 13	0.00330	0.00243	1.36
D21S11	29, 30	30, 31.2	0.00473	0.00756	0.617
D18S51	11, 19	11, 19	0.000951	1.33e-7	714
D2S441	11, 14	10, 11	0.00767	0.0125	0.614
D19S433	14, 15.2	13, 15.2	0.00700	0.00521	1.34
TH01	6, 9	6, 6	0.0118	0.00851	1.38
FGA	22, 24	22, 24	0.0219	0.00382	5.72
D22S1045	11, 15	11, 15	0.0501	0.0151	3.32
D5S818	10, 11	12, 12	0.00156	0.00624	0.250
D13S317	8, 9	8, 9	0.0249	0.00466	5.35
D7S820	10, 11	10, 11	0.0619	0.0213	2.90
SE33	19, 26.2	19, 26.2	0.00359	0.000150	23.9
D10S1248	14, 15	13, 15	0.0138	0.0171	0.806
D1S1656	14, 15	12, 14	0.000475	0.000275	1.73
D12S391	18, 19	18, 24	0.00118	0.00159	0.744
D2S1338	21, 23	23, 23	0.000170	0.0000837	2.04
Total			1.06e-50	4.27e-56	2.48e+5

Y-STR tab

Selected victim

VS

Selected reference

RS

Estimated relationship

Brother_Brother

Paternal lineage

Not excluded

Maternal lineage

Not excluded

All family members

RS

Download (STR)

Download (Y-STR)

Download (mtDNA)

STR

Y-STR

mtDNA

Locus	Victim profile	Reference profile	Ignored locus	Mismatched locus	Mutational step
DYS576	20	20	0	0	0
DYS369I	13	13	0	0	0
DYS635	21	21	0	0	0
DYS369II	29	29	0	0	0
DYS627	19	20	0	1	1
DYS460	10	10	0	0	0
DYS458	19	19	0	0	0
DYS19	15	15	0	0	0
GATA_H4	12	12	0	0	0
DYS448	18	18	0	0	0
DYS391	8	8	0	0	0
DYS456	15	15	0	0	0
DYS390	22	22	0	0	0
DYS438	13	13	0	0	0
DYS392	13	13	0	0	0
DYS518	42	42	0	0	0
DYS570	17	17	0	0	0
DYS437	13	13	0	0	0
DYS385	10,17	10,17	0	0	0
DYS449	30	30	0	0	0
DYS393	14	14	0	0	0
DYS439	12	12	0	0	0
DYS481	21	21	0	0	0
DYF367S1	37	37	0	0	0
DYS533	11	11	0	0	0
Total			0	1	1

mtDNA tab

RelSearch ver. 0.21.0 Load Result Project Database Settings Example files Manual

Result

Summary Selected data in detail Analysis conditions

Selected victim
VS
Selected reference
RS
Estimated relationship
Brother_Brother
Paternal lineage
Not excluded
Maternal lineage
Not excluded
All family members
RS

Download (STR)

Download (Y-STR)

Download (mtDNA)

STR Y-STR mtDNA

Number of mismatches
0

Shared range
73-340 16024-16365

Shared length (bp)
610

Victim profile	Reference profile	Out of shared range	Mismatch
73G	73G		
152C	152C		
263G	263G		
16071T	16071T		
16093C	16093C		
16109G	16109G		
16129A	16129A		
16223T	16223T		
16249C	16249C		
16362C	16362C		

40

5.3 Analysis conditions

You can review the following analysis conditions in **Analysis conditions** tab.

- **Database:** Input .csv file names of each database are displayed.
- **Allele probability:** Unobserved alleles in the population database are displayed, if any. You can download the allele probabilities that are estimated based on Dirichlet distribution.
- **Criteria:** The user-defined criteria to determine whether each victim-reference pair has an assumed relationship or not (see section [Criteria](#)) is displayed.
- **Assumed relationship:** Information on the user-defined relationships (see section [Relationship](#)) is displayed.
- **Mutation rate:** The user-defined mutation rates (see section [Mutation rate](#)) are displayed.
- **Data storage conditions:** The user-defined conditions for data storage (see section [Data storage conditions](#)) is displayed.

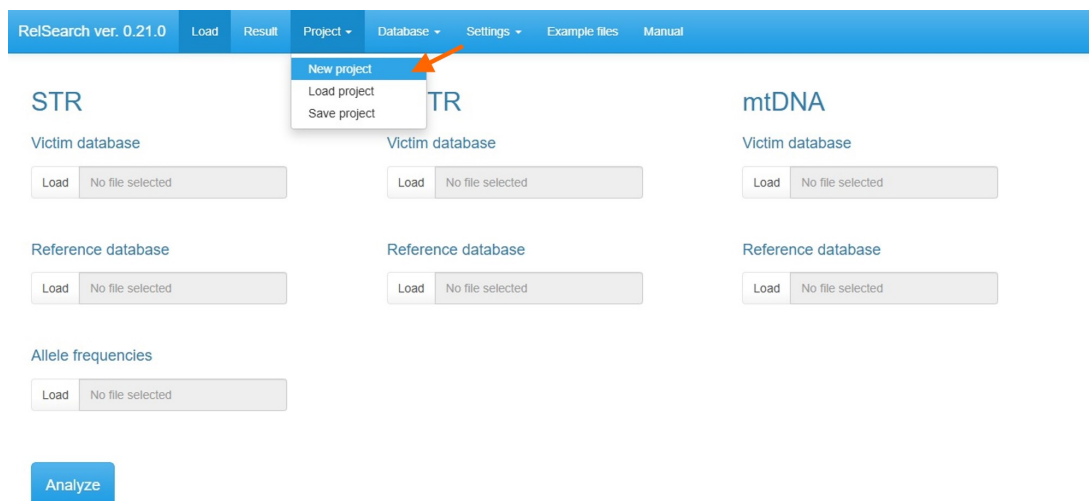
Chapter 6

Other functions

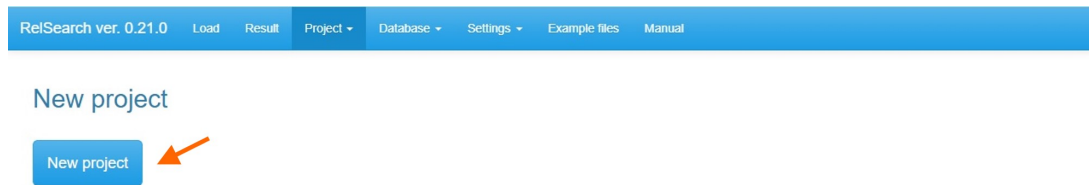
6.1 Project

6.1.1 New project

To start a new project, go to **Project ► New project**.



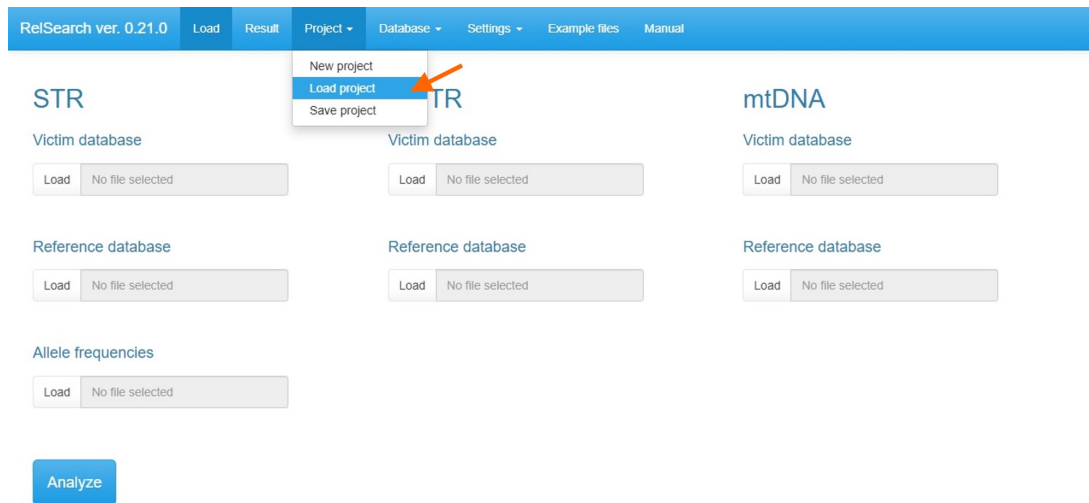
Press the **New project** button.



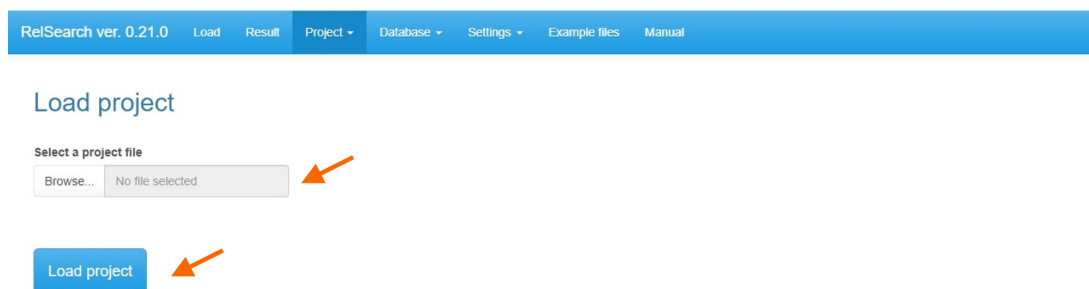
Important: The current project should be saved before starting a new project because the current project will be deleted.

6.1.2 Load project

To load a previous project, go to **Project ► Load project**.



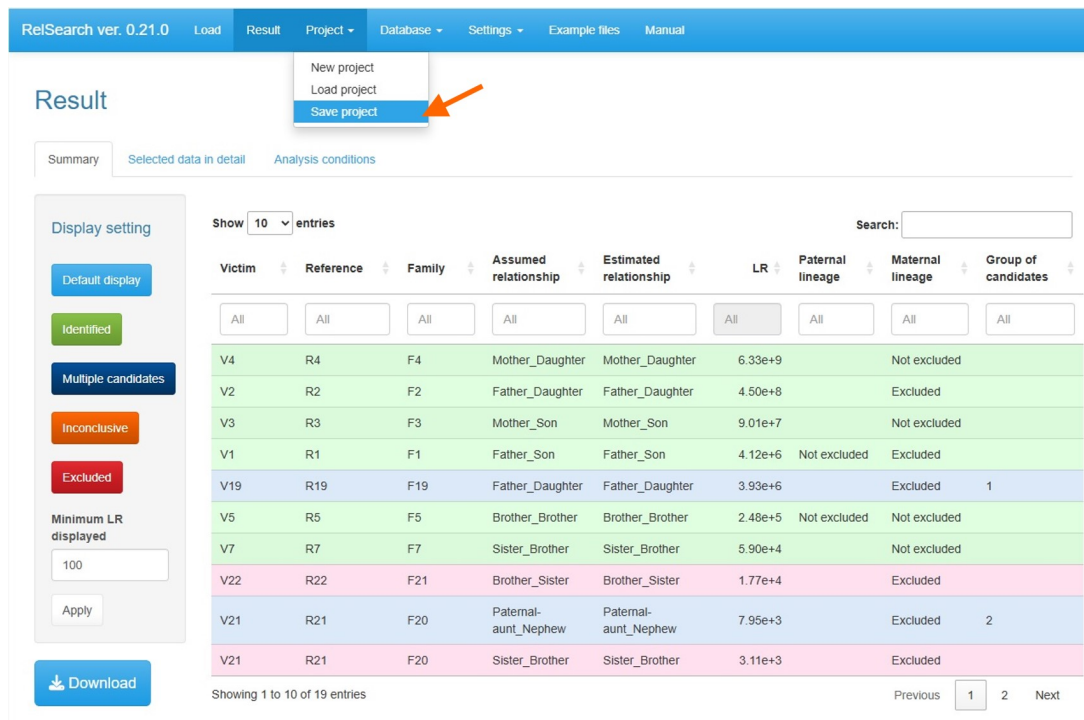
Select a project file from the **Browse...** button and press the **Load project** button.



Important: The current project should be saved before loading another project because the current project will be deleted.

6.1.3 Save project

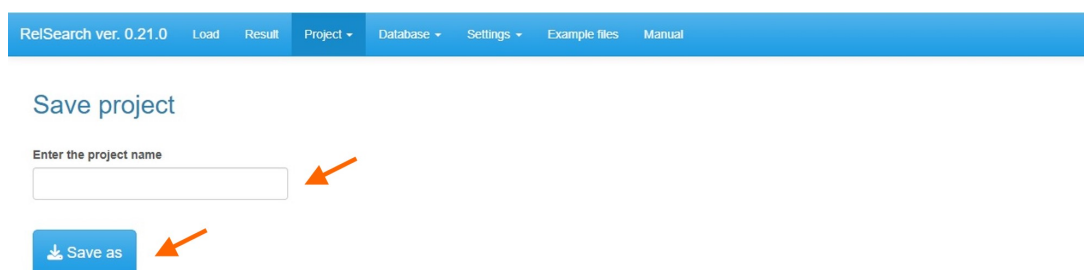
To save the current project, go to **Project ► Save project**.



The screenshot shows the RelSearch ver. 0.21.0 interface. The top navigation bar includes 'Load', 'Result', 'Project', 'Database', 'Settings', 'Example files', and 'Manual'. The 'Project' dropdown menu is open, showing 'New project', 'Load project', and 'Save project' (highlighted with an orange arrow). The 'Result' tab is active, displaying a table of search results. The table has columns: Victim, Reference, Family, Assumed relationship, Estimated relationship, LR, Paternal lineage, Maternal lineage, and Group of candidates. The table shows 19 entries, with the first 10 displayed. The 'Save project' button is located in the bottom left corner of the interface.

Victim	Reference	Family	Assumed relationship	Estimated relationship	LR	Paternal lineage	Maternal lineage	Group of candidates
V4	R4	F4	Mother_Daughter	Mother_Daughter	6.33e+9		Not excluded	
V2	R2	F2	Father_Daughter	Father_Daughter	4.50e+8		Excluded	
V3	R3	F3	Mother_Son	Mother_Son	9.01e+7		Not excluded	
V1	R1	F1	Father_Son	Father_Son	4.12e+6	Not excluded	Excluded	
V19	R19	F19	Father_Daughter	Father_Daughter	3.93e+6		Excluded	1
V5	R5	F5	Brother_Brother	Brother_Brother	2.48e+5	Not excluded	Not excluded	
V7	R7	F7	Sister_Brother	Sister_Brother	5.90e+4		Not excluded	
V22	R22	F21	Brother_Sister	Brother_Sister	1.77e+4		Excluded	
V21	R21	F20	Paternal-aunt_Nephew	Paternal-aunt_Nephew	7.95e+3		Excluded	2
V21	R21	F20	Sister_Brother	Sister_Brother	3.11e+3		Excluded	

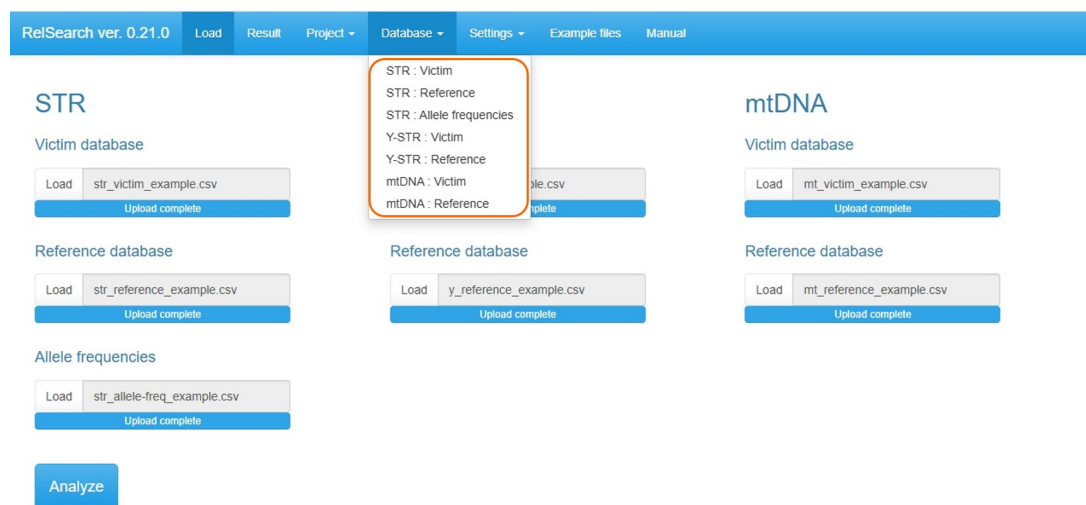
Enter the project name and press the **Save as** button.



The screenshot shows the 'Save project' dialog box in the RelSearch interface. The dialog has a text input field labeled 'Enter the project name' and a 'Save as' button. An orange arrow points to the 'Save as' button.

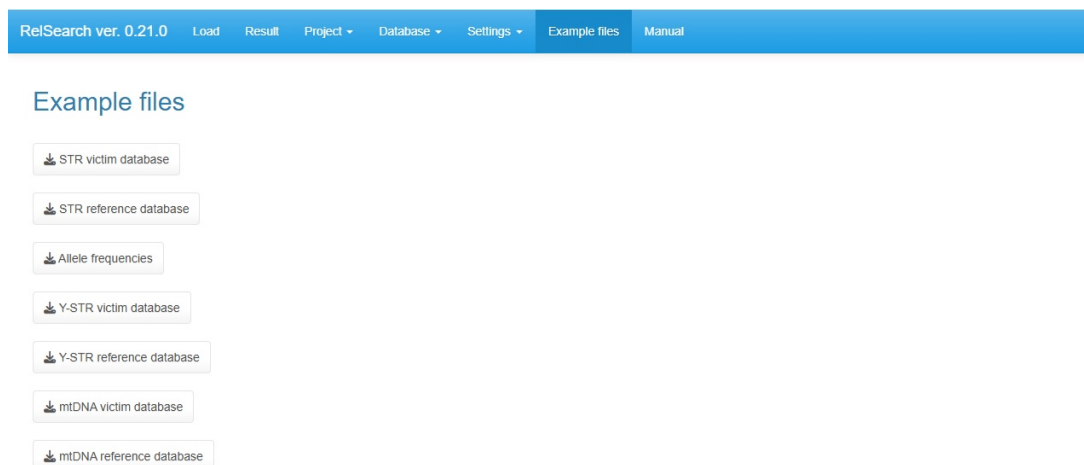
6.2 View database

Each loaded database can be reviewed in **Database** tab.



6.3 Example files

Example files for each database can be downloaded from the **Example files** tab.



References

- [1] R. M. Andrews, I. Kubacka, P. F. Chinnery, R. N. Lightowlers, D. M. Turnbull, and N. Howell. Reanalysis and revision of the Cambridge reference sequence for human mitochondrial DNA. *Nature Genetics*, 23(2):147, Oct. 1999.
- [2] C. Morimoto, H. Tsujii, S. Manabe, S. Fujimoto, E. Hirai, Y. Hamano, and K. Tamaki. Development of a software for kinship analysis considering linkage and mutation based on a Bayesian network. *Forensic Science International: Genetics*, 47:102279, July 2020.